

Assignment : A+- Troubleshooting and helpdesk

Day – 1 Tools for Troubleshooting Theory (1 hour): Overview of tools (multimeter, POST card, software diagnostics).

Task – 1

List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

➔ A multimeter is a hardware testing tool which gives output as a electrical value

Real Life Uses of a Multimeter

Scenarios	How do Multimeter uses
Computer Power Supply Problem	Multimeter is use to check PSU output whether is giving correct voltage
Adapter Cable problem	Cchecks adapter is correct output and cable voltage
Damaged or broken Cable	Can check Cable continuity

Example 1 – PSU

Suppose a desktop suddenly doesn't turn on.

Instead of immediately replacing the PSU, a technician can use a multimeter to check whether the PSU is producing the expected voltage.

Example 2 – Laptop charger

Laptop isn't charging.

The technician can check the adapter's output voltage using a multimeter. If the output is abnormal, the charger may be faulty.

Example 3 – Cable

Suppose a cable looks fine from outside but isn't working.

Using **continuity mode**, the technician can check whether there is an electrical connection through the cable.

Task – 2

Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

Hint: Focus on the codes shown and how they relate to hardware issues.



Post CARD is Connected to the MOTHERBOARD , than Computer is Power ON , CARD LED Light becomes Active and Different codes appears to diagnosis the Problem

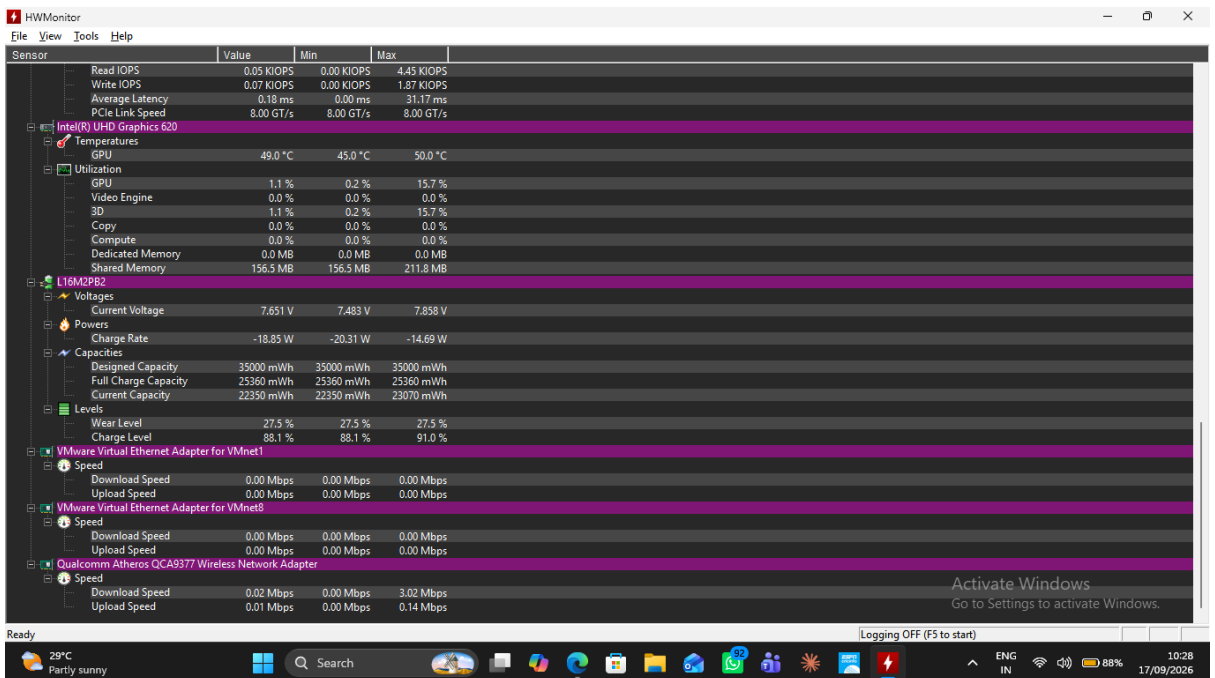
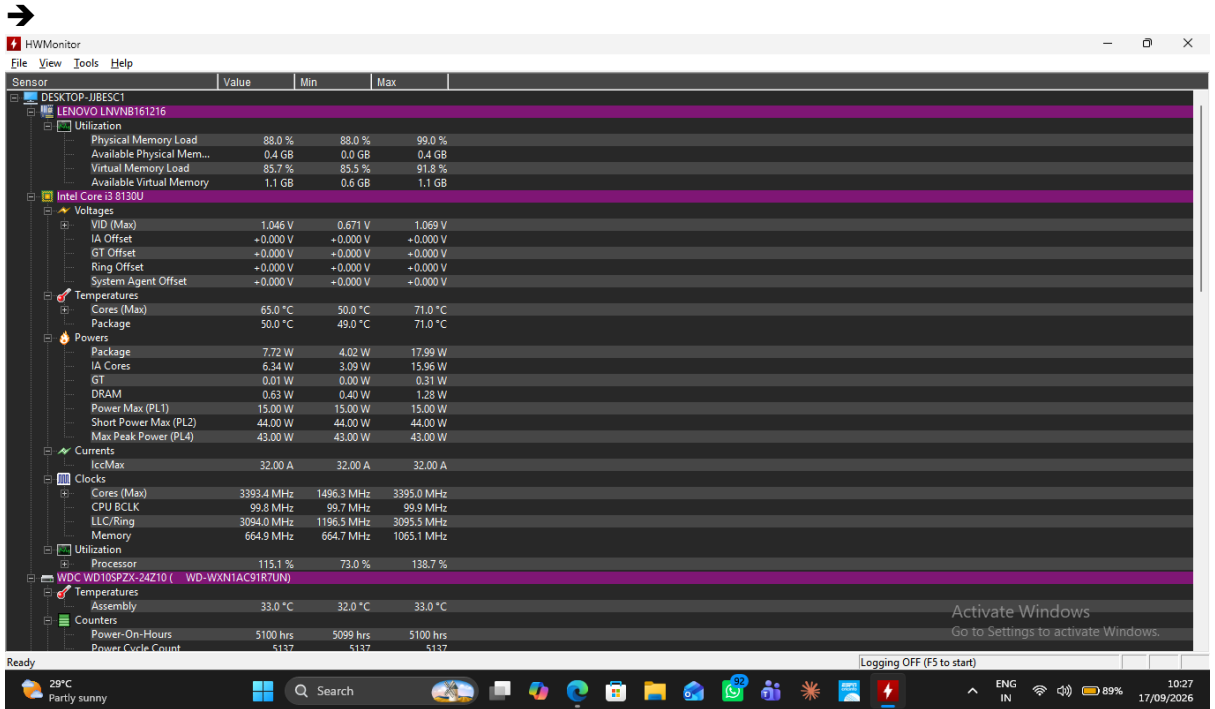
A D4 Code which is a POST CODE diagnosis code it indicates a problem while the BIOS is detecting or initializing a Hardware Exact meaning depends on BIOS / MOTHERBOARD.

A D5 Code it indicates a problem when motherboard has reached a particular hardware intialization .

Task - 3

Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.

Constraint: Blur or crop out any personal information before submitting.



Task – 4

Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.



TOOLS	WHAT IT CHECKS	EXAMPLE PROBLEM DETECTED
MULTIMETER	Checks Output Voltage , Current , Resistance , Continuity	When any damaged cable , PSU, adapter charging problem
POST CARD	Monitors the POST process and give diagnosis code during startup	RAM , CPU ,ROM , MOTHERBOARD and hardware related problem
SOFTWARE DIAGNOSTICS	Checks hardware temperature and fan and sensors	Overheating CPU , GPU and failing sensors